Chapter-1.

1. List the first 16 numbers in base 12. Use the letters A and B to represent the last two digits.

Ans:

0, 1, 2, 3, 4, 5, 7, 8, 9,

A, B, 10, 11, 12, 13, 14

2. What is the largest binary number that can be obtained with 16 bits? What is its decimal equivalent?

Ans:

The Largest binary number that can be obtained with 16 bits is

1111111111111111

It's decimal equivalent is 65535.

3. Convert the following binary numbers to decimal: 101110, 1110101.11 and 110110100.

Ans:

$$(101110)_2 = 1.2^5 + \cancel{1.2^4}\ 0.2^4 + 1.2^3 + 1.2^2 + 1.2^1 + 0.2^0$$

$$= (46)_{10}$$

$$(1110101.11)_2 = 1\times2^6 + 1\times2^5 + 1\times2^4 + 0\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0 + 1\times2^{-1} + 2^{-2}\times1$$

$$= (117.75)_{10}$$

$$(110110100)_2 = 1\times2^8 + 1\times2^7 + 0\times2^6 + 1\times2^5 + 1\times2^4 + 0\times2^3 + 1\times2^2 + 0\times2^1 + 0\times2^0$$

$$= (436)_{10}$$

4. convert the following numbers with the indicated bases to decimal:

$(12121)_3$, $(4310)_5$, $(50)_7$, $(198)_{12}$

Ans:

$$(12121)_3 = 1\times3^4+2\times3^3+1\times3^2+2\times3^1+1\times3^0$$

$$= (151)_{10}$$

$$(4310)_5 = 4\times5^3+3\times5^2+1\times5^1+0\times5^0$$

$$= (580)_{10}$$

$$(50)_7 = 5\times7^1+0\times7^0$$

$$= (35)_{10}$$

$$(198)_{12} = 1\times12^2+9\times12^1+8\times12^0$$

$$= (260)_{10}$$

5. convert the following decimal numbers to binary : 1231, 673.23, 10^4, 1998.

Ans :

$(1231)_{10}$ =

2	1231
2	615 – 1
2	307 – 1
2	153 – 1
2	76 – 1
2	38 – 0
2	19 – 0
2	9 – 1
2	4 – 1
2	2 – 0
2	1 – 0
	0 – 1

$(1231)_{10} = (1011\ 01111)_2$

$(673.23)_{10} = (1010100001.0011101011)_2$

$(10^4)_{10} = (10011100010000)_2$

$(1998)_{10} = (11111001110)_2$

6. Convert the following decimal numbers to the indicated bases:

a) 7562.45 to octal

Ans:

$(7562.45)_{10} =$

$$
\begin{array}{r|l}
8 & 7562 \\
8 & 945 - 2 \\
8 & 118 - 1 \\
8 & 14 - 6 \\
8 & 1 - 6 \\
 & 0 - 1
\end{array}
$$

	int		fraction
$.45 \times 8 =$	3	+	.6
$.6 \times 8 =$	4	+	.8
$.8 \times 8 =$	6	+	.4
$.4 \times 8 =$	3	+	.2
$.2 \times 8 =$	1	+	.6
$.6 \times 8 =$	4	+	.8

$\therefore (7562.45)_{10} = (16612.3463146314)_8$

b) ~~Binary~~ 1938.257 to hexadecimal

Ans:

$$\begin{array}{r|l} 16 & 1938 \\ \hline 16 & 121 - 2 \\ \hline 16 & 7 - 9 \\ \hline & 0 - 7 \end{array}$$

$.257 \times 16 = 4 + .112$

$.112 \times 16 = 1 + .792$

$.792 \times 16 = c\ 12 + .672$

$(1938 \cdot 257)_{10} = (792 \cdot 41CAC)_{16}$

c) 175·175 to binary

Ans:

$(175 \cdot 175)_{10} = (10101111 \cdot 001011011)_2$

7. convert the hexadecimal numbere F3A7C2 to binary and octal

Ans:

B/.

$$\begin{array}{cccccc} F & 3 & A & 7 & C & 2 \\ 1111 & 0011 & 1010 & 0111 & 1100 & 0010 \end{array}$$

$(1111\ 0011\ 1010\ 0111\ 1100\ 0010)_2$

F 3 A 7 C ~~6~~ 2

1111 0011 1010 0111 1100 0010

$\underbrace{111}_{7}\ \underbrace{100}_{4}\ \underbrace{111}_{7}\ \underbrace{010}_{2}\ \underbrace{011}_{3}\ \underbrace{111}_{7}\ \underbrace{000}_{0}\ \underbrace{010}_{2}$

$(F3A7C2)_{16} = (74723702)_8$

8. Convert the following numbers from the given base to other three bases.

a) Decimal 225 to binary, octal, hexadecimal.

Ans:

$(225)_{10} = (11100001)_2$

$(225)_{10} = (341)_8$

$(225)_{10} = (E1)_{16}$

b) Binary 11010111 to decimal, octal, hexadecimal.

Ans:

$(11010111)_2 = (215)_{10}$

$(11010111)_2 = (327)_8$

$(11010111)_2 = (D7)_{16}$

c) Octal 623 to decimal, binary, hexadecimal.

Ans:

$(623)_8 = (403)_{10}$

$(623)_8 = (110010011)_2$

$(623)_8 = (193)_{16}$

d) Hexadecimal 2AC5 to decimal octal, binary.

Ans:

$(2AC5)_{16} = (10949)_{10}$

$(2AC5)_{16} = (25305)_8$

$(2AC5)_{16} = (10101011000101)_2$

9. Add and multiply the following number without converting decimal

a) $(367)_8$ and $(715)_8$

$(367)_8 = 11110111$

$(715)_8 = 111001101$

$(1304)_8 \quad 1011000100$

$$\begin{array}{r} 367 \\ 715 \\ \hline 2323 \\ 367\times \\ 3301\times \\ \hline (336313)_8 \end{array}$$

b) $(15F)_{16}$ and $(A7)_{16}$

$$\begin{array}{lcr} 15F & = & 101011111 \\ A7 & = & 10100111 \\ \hline (206)_{16} & & 1000000110 \end{array}$$

$$\begin{array}{r} 15F \\ A7 \\ \hline E4F9 \end{array}$$

c) $(110110)_2$ and $(110101)_2$

$$\begin{array}{r} 110110 \\ +\ 110101 \\ \hline 61101011 \end{array}$$

$$\begin{array}{r} 110110 \\ \times\ 110101 \\ \hline 0101100101110 \end{array}$$

10. Perform the following division in binary:

11111111/101

11111111 = 255

101 = 5

$$5\,|\,255\,|\,51$$
$$\begin{array}{r} 25 \\ \hline 25 \\ 25 \\ \hline \times \end{array}$$

51 = 0110011

11. Determine the value of base x if $(211)_x = (152)_8$

$(211)_x = (152)_8$

$2 \times x^2 + 1 \times x^1 + 1 \times x^0 = 1 \times 8^2 + 5 \times 8^1 + 2 \times 8^0$

$\Rightarrow 2x^2 + x + 1 = 64 + 40 + 2$

$\Rightarrow 2x^2 + x + 1 = 106$

$\Rightarrow 2x^2 + x + 1 - 106 = 0$

$\Rightarrow 2x^2 + x - 105 = 0$

$x_1 = 7$

$x_2 = -7.5$

$\therefore x = 7$

12. Noting that $3^2 = 9$, formulate a simple procedure for converting base 3 numbers directly to base-9. Use the procedure to convert $(2110.20110222 0112)_3$ to base 9.

Ans!

$$3^2 = 9$$

All the value below 3^2 has to be accomodated in 9^0 which also means all values in 9^0 have to be accomodated in the two places below 3^2

This rule also applies to conversion between binary, octal or hexadecimal

In general for conversion N in any base k to base k^n each digit of N_{k^n} get converted to n digits of N_k

$\therefore (2110201102220112)_3$

$= 2\times3^{15}+1\times3^{14}+1\times3^{13}+0\times3^{12}+2\times3^{11}+0\times3^{10}+1\times3^{9}+1\times3^{8}+$
$0\times3^{7}+2\times3^{6}+2\times3^{5}+2\times3^{4}+0\times3^{3}+1\times3^{2}+1\times3^{1}+2\times3^{0}$

$= (73642815)_9$

13. Find the 9's complement of the following 8-digit decimal numbers. 12349876; 00980100; 90000951; 00000000

Ans:
$$\begin{array}{r} 99999999 \\ -\ 12349876 \\ \hline 87650123 \end{array}$$

$$\begin{array}{r} 99999999 \\ -\ 00980100 \\ \hline 99019899 \end{array}$$

$$\begin{array}{r} 99999999 \\ -90009951 \\ \hline 09990048 \end{array} \qquad \begin{array}{r} 99999999 \\ -00000000 \\ \hline 99999999 \end{array}$$

14. Find 10's complement of the following (6 digits): 123900, 090657, 100000, 000000

Ans:

i)
$$\begin{array}{r} 999999 \\ -123900 \\ \hline 876099 \end{array} \qquad \begin{array}{r} 876099 \\ +1 \\ \hline 876100 \end{array}$$

ii)
$$\begin{array}{r} 999999 \\ -090657 \\ \hline 909342 \\ +1 \\ \hline 909343 \end{array}$$

iii)
$$\begin{array}{r} 999999 \\ -100000 \\ \hline 899999 \\ +1 \\ \hline 900000 \end{array}$$

iv) $$\begin{array}{r} 999999 \\ -000000 \\ \hline 999999 \\ +1 \\ \hline \cancel{1}000000 \end{array}$$

15. Find 1's and 2's complement of the following 8 digit binary: 10101110, 10000001, 10000000, 00000000

Ans: 1's complement

$$\begin{array}{r} 11111111 \\ -10101110 \\ \hline 01010001 \end{array}$$

2's complement

$$\begin{array}{r} 11111111 \\ -10101110 \\ \hline 01010001 \\ +1 \\ \hline 01010010 \end{array}$$

$$\begin{array}{r} 11111111 \\ -10000001 \\ \hline 01111110 \end{array}$$

$$\begin{array}{r} 11111111 \\ -10000001 \\ \hline 01111110 \\ +1 \\ \hline 01111111 \end{array}$$

$$\begin{array}{r} 11111111 \\ -10000000 \\ \hline 01111111 \end{array}$$

$$\begin{array}{r} 11111111 \\ -10000000 \\ \hline 01111111 \\ +1 \\ \hline 10000000 \end{array}$$

$$\begin{array}{r} 11111111 \\ -00000000 \\ \hline 11111111 \end{array}$$

$$\begin{array}{r} 11111111 \\ -00000000 \\ \hline 11111111 \\ +1 \\ \hline 00000000 \end{array}$$

16. Perform substraction with the following unsigned decimal numbers by taking the 10's complement of the substracted.

a) 5250 − 1321

10's complement of 1321

$$\begin{array}{r} 9999 \\ -1321 \\ \hline 8678 \\ +1 \\ \hline 8679 \end{array}$$

10's — 8679

$$\begin{array}{r} 5250 \\ +\ 8679 \\ \hline \text{carry} \leftarrow \cancel{1}\,3929 \end{array}$$

carry থাকলে 2nd num comp কারে + করে ⇒ ans যেটা আসবে সেটা

b) 1753 − 8640

$$\begin{array}{r} 1753 \\ -\ 8640 \\ \hline -6887 \end{array} \xrightarrow{10's} \begin{array}{r} 1753 \\ +1360 \\ \hline 3113 \end{array} \quad -6887 \xleftarrow{10's} 3113$$

⇒ carry না থাকলে 2nd num এর comp কারে + কারে ans এর carry না থাকলে ans এর comp কারে ans

c) 20 − 100

$$\begin{array}{r} 020 \\ -\ 100 \\ \hline -\ 080 \end{array} \xrightarrow{10's} \begin{array}{r} 020 \\ +900 \\ \hline 920 \end{array} \quad -080 \xleftarrow{10's} 920$$

d) 1200 − 250

$$\begin{array}{r} 1200 \\ -0250 \\ \hline \end{array} \xrightarrow{10's} \begin{array}{r} 1200 \\ +9750 \\ \hline \underset{\text{carry}}{\cancel{1}}0950 \end{array}$$

17. Perform the substraction with the following unsigned binary by taking 2's complement

Ans:

a) 11010 − 10000

$$\begin{array}{r} 11010 \\ -\,10000 \\ \hline 01010 \end{array} \quad \xrightarrow{2's} \quad \begin{array}{r} 11010 \\ +\,10000 \\ \hline 1\,01010 \end{array}$$

1 = carry

b) 11010 − 1101

$$\begin{array}{r} 11010 \\ -\,01101 \\ \hline 01101 \end{array} \quad \xrightarrow{2's} \quad \begin{array}{r} 11010 \\ +\,10011 \\ \hline 1\,01101 \end{array}$$

1 = carry

c) 100 − 110000

$$\begin{array}{r} 000100 \\ -\,110000 \\ \hline 101100 \end{array} \quad \xrightarrow{2's} \quad \begin{array}{r} 000100 \\ +\,010000 \\ \hline 010100 \end{array} \quad \xrightarrow{2's} \quad 101100$$

d) $1010100 - 1010100$

$$\begin{array}{r} 1010100 \\ -\ 1010100 \\ \hline 0000000 \end{array} \xrightarrow{2's} \begin{array}{r} 1010100 \\ +0101100 \\ \hline \underset{carry}{1}\,0000000 \end{array}$$

18. Perform the arithmetic operation $(+42)+(-13)$ and $(-42)-(-13)$ in binary using the signed 2's complement representation for negative number.

Ans:

$+42 = 0101010$

$13 = 01101$

$-42 = 1010110$

$-13 = \cancel{10011}\ 1110011$

$42+(-13) = 0101010 + 1110011$

$= 0011101$

$= (29)_{10}$

$(-42)-(-13) = (-42)+13 = 1010110 + 0001101$

$= 1100011$

$= (-11101)_2 = (-29)_{10}$

19. The binary numbers listed have a sign in the leftmost position and if negative, are in 2's complement form. Perform arithmetic operation.

a) 101011 + 111000

Ans) 101011 = -21

111000 = -8

$$\begin{array}{r} -21 \\ -8 \\ \hline -29 \end{array}$$

(-29) = 1100011

b) 001110 + 110010

001110 = 14

110010 = -14

$$\begin{array}{r} 14 \\ -14 \\ \hline 0 \end{array}$$

0 = 0.000000

c) 111001 - 001010

111001 = -7

001010 = 10

$$\begin{array}{r} -7 \\ -10 \\ \hline -17 \end{array}$$

(-17) = 101111

d) 101011 − 100110

101011 = (−21)

100110 = −26

$$-21 - (-26) \qquad \begin{array}{r} -21 \\ +26 \\ \hline 5 \end{array}$$

000101 = 5

20. Represent the following decimal numbers in BCD: 13597, 93286, 99880

Ans:

1 = 0001

3 = 0011

5 = 0101

9 = 1001

7 = 0111

13597 = 00010011010110010111

93286

9 = 1001 8 = 1000

3 = 0011 6 = 0110

2 = 0010

93286 = 1001 0011 0010 1000 0110

99880

9 = 1001

9 = 1001

8 = 1000

8 = 1000

0 = 0000

99880 = 1001 1001 1000 1000 0000

21. Determine the binary for each of the ten decimal digits using weighted code with weights 7, 4, 2, 1

Ans:

Decimal Digit	7421 code
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	1000
8	1001
9	1010

22. The (r-1)'s complement of base-6 numbers is called the 5's complement

a) Determine a procedure for obtaining the 5's complement of base-6 numbers.

Ans:

* First, substract 5's from the given number.
* Then add 1 to the resultant.
* This will give us 5's complement of base-6 numbers

b) Obtain the 5's complement of $(543210)_6$

Ans:

$$\begin{array}{r} 555555 \\ -\ 543210 \\ \hline 12345 \\ +1 \\ \hline 12346 \end{array}$$

c. Design a 3bit binary code to represent each of the six digits of the base-6 number system. make the binary code self complementing.

Ans:

212 code is self complementing code for base 6 as sum = 5

23. Represent decimal number 8620 in

a) BCD

Ans:

8 = 1000

6 = 0110

2 = 0010

0 = 0000

8620 = 1000 0110 0010 0000

b) excess - 3 code

Ans:

$8 + 3 = 11$ — 1011

$6 + 3 = 9$ — 1001

$2 + 3 = 5$ — 0101

$0 + 3 = 3$ — 0011

$(8620)_{10} = (1011100101010011)_{S3}$

c) 2421 code

Ans:

8 = 1110

6 = ~~0100~~ 1100

2 = 0010

0 = 0000

$(8620)_{10} = (1110\ 11000010\ 0000)_{2421\ code}$

d) as a binary number:

$(8620)_{10} = (10000110101100)_2$

24. Represent decimal 3864 in the 2421 code of table-1-2. show that the code is self complementing by taking 9's complement of 3864.

Ans:

3 = 0011

8 = 1110

6 = 1100

4 = 0100

(3864) = 0011 1110 1100 0100

9's complement of 3864 is 6135

which in 2421 is 1100 0001 0011 1011

(0011111011000100) = 3864

(1100000100111011) = (6135)

compare the two, we get each other by interchanging 0's and 1's self complementing codes provide the 9's complement of a decimal number, just by interchanging 1's and 0's in its equivalent 2421 representation.

25. Assign a binary code in some orderly manner to the 52 playing cards. Use the minimum number of bits.

Ans:

$$2^x = 52$$

$$x = 5.7$$

$$x = 6$$

$\therefore$ 6 bits required to represent 52 cards.

Ace of spade = 000000
" " Diamond = 001101

largest card = 110011

26. List the ten BCD digits with an even parity in the leftmost position. Repeat with an odd parity bit.

Ans:

Even parity: 00000, 10001, 10010, 00011, 10100, 00101, 00110, 10111, 11000, 01001

Odd parity: 10000, 00001, 00010, 10011, 00100, 10101, 10110, 00111, 01000, 11001.

27. Write your full name in ASCII using an eight bit code with the leftmost bit always 0.

Ans:

John F. Kennady - 01001010 01101111 01101000 01101110 00100000 , 01000110 00101110 , 01001011 , 01100101 01101110 01101110 01101110 01100001 01100100 01111001

28. Decode the following ASCII code: 1001010 1101111 1101000 1101110 0100000 1000100 1101111 1100101

Ans:

J o h n Doe

30. How many printing characters are there in ASCII? How many of them are not letters or numerals?

Ans: There are 95 printable ASCII characters. 0 - 47, 58 to 64, 91 to 96 and 123 to 127 are not letters or numerals.

31. The state of a 12 bit register is 010110010111. What is its content if it represents:

a) three decimal digits in BCD.

Ans:

$\underbrace{010}_{5}\,\underbrace{1\,1001}_{9}\,\underbrace{0111}_{7}$

BCD = 597

b) three decimal digits in the excess-3 code

Ans) 0 8 11 10

c) three decimal digits in the 2421

Ans: 5 3 7

32. Show the contents of all r
in fig -13 if the two binary
numbers added have the dec
257 and 514.

Ans:

Binary conversion for given two
decimal 257 and 514:

257 = 0100000001

514 = 1000000010

convert to binary and then take fir
10 LSB's. Because all registers in abo
system are 10-bit registers.

$R_1 = 0100000001$

$R_2 = 1000000010$

Finally R_3 stores sum of the values
in R_1 and R_2 i.e. $R_3 <= R_1 + R_2$

$R_3 = 1100000011$

	x	y	
0	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

33. Show the signals of the outputs F and G in the two gates of fig-1-6 (d) and (e). Use all 16 possible combinations of the input signals. A, B, C, D.

Ans:

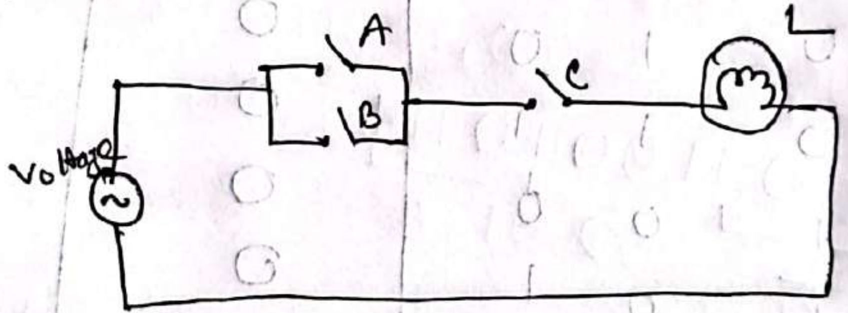
A	B	C	D	A B C	A+B+C+D
0	0	0	0	0	0
0	0	0	1	0	1
0	0	1	0	0	1
0	0	1	1	0	1
0	1	0	0	0	1
0	1	0	1	0	1
0	1	1	0	0	1
0	1	1	1	0	1
1	0	0	0	0	1
1	0	0	1	0	1
1	0	1	0	0	1
1	0	1	1	0	1
1	1	0	0	0	1
1	1	0	1	0	1
1	1	1	0	1	1
1	1	1	1	1	1

signal for F and G

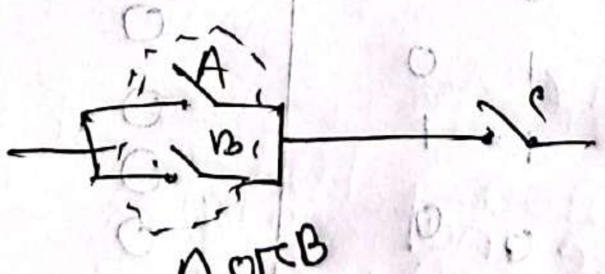
F 0000000000000000 1 1

G 0 1111111111111111

34. Express the switching circuit show in the figure in binary notation.



Ans:



$F(A, B, C) = (A+B)\,C$

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1